ASSIGNMENT 4-5 REPORT

Incident Response Simulation and
Infrastructure as Code Implementation

MicroShop - Containerized Microservices System

Introduction to SRE | 2026

GitHub: https://github.com/Nurlybek-Nurzhan/Assignment_4-5_SRE

System Overview

MicroShop is an e-commerce application deployed in a containerized environment designed for illustrating Site Reliability Engineering techniques. The MicroShop application is composed of five separate FastAPI microservices, a PostgreSQL database, an Nginx web server operating as a reverse proxy, Prometheus monitoring system configured for data gathering and alerting, and Grafana for automated dashboard generation.

Architecture

- Auth Service (port 8001) - JWT authentication with bcrypt password hashing; credentials from environment variables
- Product Service (port 8002) - product catalog, persisted to a Docker volume via JSON file
- Order Service (port 8003) - order management, PostgreSQL backend; this is the incident target
- User Service (port 8004) - user profile management
- Chat Service (port 8005) - messaging with JWT token verification on all endpoints
- PostgreSQL - persistent database, used exclusively by the Order Service
- Prometheus (port 9090) - scrapes /metrics from all five services; alert rules defined for error detection
- Grafana (port 3000) - auto-provisioned datasource and dashboard on startup
- Nginx (port 80) - serves the frontend, proxies /api/* routes to backend services

Security Hardening Applied

- Auth credentials stored in .env file, never hardcoded in source code
- Passwords hashed with bcrypt - plaintext passwords never stored
- Real JWT tokens (HS256) issued on login; Chat Service validates token on every request
- All containers have memory and CPU resource limits (noisy-neighbor prevention)
- Docker healthchecks defined for all FastAPI services and Nginx
- terraform.tfstate excluded from git via .gitignore - contains sensitive AWS resource IDs

Known Limitations (Assignment Scope)

- Auth service uses two hardcoded users defined in .env - no user registration flow
- Chat messages are in-memory only - reset on container restart (no chat DB)
- Security group opens ports to 0.0.0.0/0 by default - the allowed_ssh_cidr variable allows restricting to a specific IP
- No SSH key pair configured for the EC2 instance - the ssh_key_name variable is available but left empty for this demo

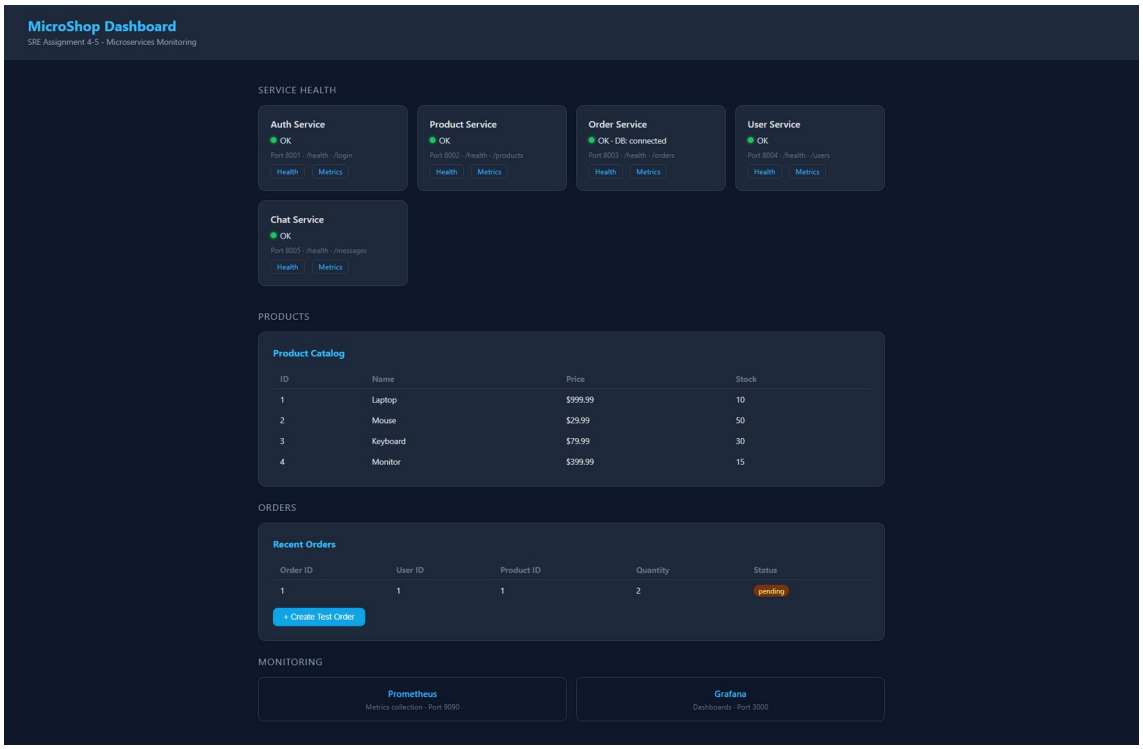
Running System

```
PROBLEMS   DEBUG CONSOLE   OUTPUT   TERMINAL   PORTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
time="2026-04-29T16:10:58+05:00" level=warning msg="C:\\Users\\Nurzhan\\Favorites\\Basa\\Education\\Programming\\Introduction to SRE\\assignment-4-5\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, pl
ease remove it to avoid potential confusion"
[+] up 10/10
 ✓ Network assignment-4-5_microshop-net Created                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.1s
 ✓ Container postgres                   Created                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.8s
 ✓ Container prometheus                 Created                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.8s
 ✓ Container grafana                    Created                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.8s
 ✓ Container user-service               Created                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.7s
 ✓ Container frontend                   Created                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.8s
 ✓ Container product-service            Created                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.6s
 ✓ Container auth-service               Created                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.6s
 ✓ Container chat-service               Created                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.7s
 ✓ Container order-service              Created                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.5s
Attaching to auth-service, chat-service, frontend, grafana, order-service, postgres, product-service, prometheus, user-service
postgres  |
postgres  | PostgreSQL Database directory appears to contain a database; Skipping initialization
postgres  |
postgres  | 2026-04-29 11:11:01.466 UTC [1] LOG:  starting PostgreSQL 15.17 on x86_64-pc-linux-musl, compiled by gcc (Alpine 15.2.0) 15.2.0, 64-bit
postgres  | 2026-04-29 11:11:01.466 UTC [1] LOG:  listening on IPv4 address "0.0.0.0", port 5432
postgres  | 2026-04-29 11:11:01.466 UTC [1] LOG:  listening on IPv6 address "::", port 5432
postgres  | 2026-04-29 11:11:01.496 UTC [1] LOG:  listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
postgres  | 2026-04-29 11:11:01.540 UTC [29] LOG:  database system was shut down at 2026-04-29 11:10:44 UTC
postgres  | 2026-04-29 11:11:01.569 UTC [1] LOG:  database system is ready to accept connections
user-service  | INFO:     Started server process [1]
user-service  | INFO:     Waiting for application startup.
user-service  | INFO:     Application startup complete.
user-service  | INFO:     Uvicorn running on http://0.0.0.0:8004 (Press CTRL+C to quit)
product-service  | INFO:     Started server process [1]
product-service  | INFO:     Waiting for application startup.
product-service  | INFO:     Application startup complete.
product-service  | INFO:     Uvicorn running on http://0.0.0.0:8002 (Press CTRL+C to quit)
auth-service     | INFO:     Started server process [1]
auth-service     | INFO:     Waiting for application startup.



auth-service     | INFO:     Uvicorn running on http://0.0.0.0:8001 (Press CTRL+C to quit)

chat-service     | INFO:     Application startup complete.
chat-service     | INFO:     Uvicorn running on http://0.0.0.0:8005 (Press CTRL+C to quit)
Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint grafana (b4a0f305c16adaa710896e265da9eb0c4e53a3b6a9f0392acab75071d368c021): failed to bind host port 0
.0.0.0:3000/tcp: address already in use
v View in Docker Desktop   o View Config   w Enable Watch   d Detach
Nurzhan@Nurzhan MINGW64 ~/Favorites/Basa/Education/Programming/Introduction to SRE/assignment-4-5
$
```

```
docker compose up -d
time="2026-04-29T16:42:16+05:00" level=warning msg="C:\\Users\\Nurzhan\\Favorites\\Basa\\Education\\Programming\\Introduction to SRE\\assignment-4-5\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, pl
ease remove it to avoid potential confusion"
[+] up 9/9
 ✓ Container chat-service    Started                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                2.1ss
 ✓ Container postgres        Healthy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                12.5s
 ✓ Container user-service    Started                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                2.2ss
 ✓ Container grafana         Started                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                2.2ss
 ✓ Container prometheus      Started                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                2.2ss
 ✓ Container frontend        Started                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                1.9ss
 ✓ Container product-service Started                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                2.0ss
 ✓ Container auth-service    Started                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                1.9ss
 ✓ Container order-service   Started                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                12.4s

Nurzhan@Nurzhan MINGW64 ~/Favorites/Basa/Education/Programming/Introduction to SRE/assignment-4-5
$ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED          STATUS                    PORTS                                          NAMES
063f93add541   assignment-4-5-order-service   "uvicorn main:app --…"   24 seconds ago   Up 11 seconds             0.0.0.0:8003->8003/tcp, [::]:8003->8003/tcp    order-service
2a823a574beb   prom/prometheus:latest         "/bin/prometheus --c…"   25 seconds ago   Up 22 seconds             0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp    prometheus
249a23b67477   assignment-4-5-user-service    "uvicorn main:app --…"   25 seconds ago   Up 22 seconds             0.0.0.0:8004->8004/tcp, [::]:8004->8004/tcp    user-service
91a185d11b6a   nginx:alpine                   "/docker-entrypoint.…"   25 seconds ago   Up 22 seconds             0.0.0.0:80->80/tcp, [::]:80->80/tcp            frontend
3674f2d5f94f   postgres:15-alpine             "docker-entrypoint.s…"   25 seconds ago   Up 22 seconds (healthy)   5432/tcp                                       postgres
38b12b3b6dec   assignment-4-5-auth-service    "uvicorn main:app --…"   25 seconds ago   Up 22 seconds             0.0.0.0:8001->8001/tcp, [::]:8001->8001/tcp    auth-service
22da1596bf20   assignment-4-5-product-service "uvicorn main:app --…"   25 seconds ago   Up 22 seconds             0.0.0.0:8002->8002/tcp, [::]:8002->8002/tcp    product-service
575105dd5922   grafana/grafana:latest         "/run.sh"                25 seconds ago   Up 22 seconds             0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp    grafana
0be8f4eda280   assignment-4-5-chat-service    "uvicorn main:app --…"   25 seconds ago   Up 22 seconds             0.0.0.0:8005->8005/tcp, [::]:8005->8005/tcp    chat-service
```

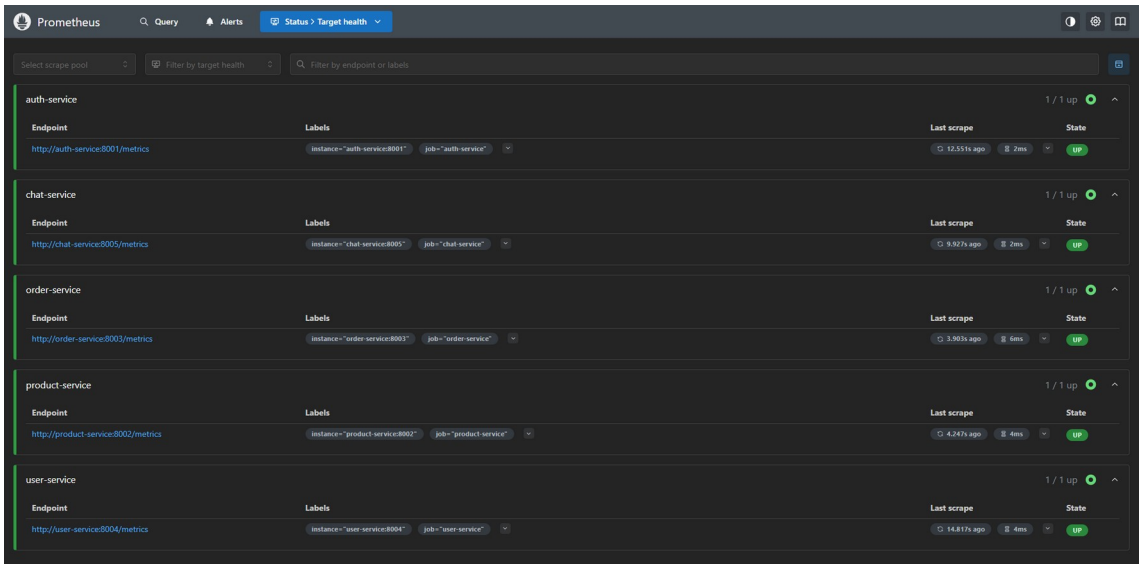
Frontend Dashboard

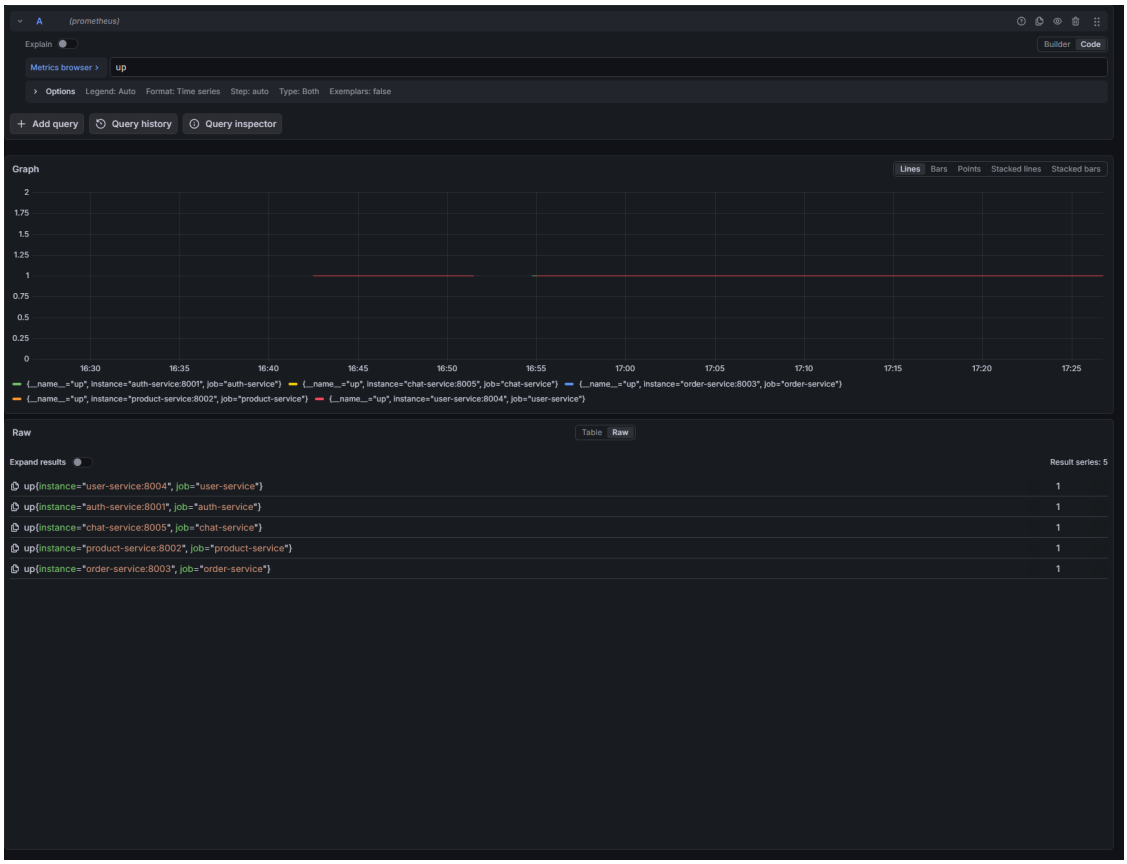
The MicroShop Dashboard shows real-time health of all five services with green/red status dots. The header includes a login form - entering credentials calls the Auth Service and receives a JWT token which is then used for authenticated Chat Service calls.



Monitoring

Prometheus scrapes all five services at their /metrics endpoints every 15 seconds. Two alert rules are defined in alert_rules.yml: OrderServiceErrors (fires when order_errors_total increases) and ServiceDown (fires when any target becomes unreachable). Grafana loads the Prometheus datasource and the MicroShop dashboard automatically on startup via provisioning files - no manual configuration needed.





ASSIGNMENT 4 - Incident Report

1. Incident Summary

On 2026-04-29, a production-level incident was simulated in the MicroShop Order Service by deliberately misconfiguring the DB_HOST environment variable from 'postgres' to 'wrong-host'. After restarting the container with this broken configuration, the Order Service lost all database connectivity and returned HTTP 503 on every request. All four other services remained fully operational throughout the incident, demonstrating correct fault isolation.

2. Impact Assessment

- Order Service: 100% unavailable - /orders (GET and POST) and /health all returned HTTP 503
- Frontend dashboard: Order Service card showed 'Unreachable' with red status dot
- Users could not view existing orders or create new ones
- Auth, Product, User, and Chat services: fully operational - fault isolation confirmed
- Data integrity: no data was lost - PostgreSQL remained healthy throughout the incident
- Prometheus showed order-service as up=1 - the process was running but functionally broken

3. Severity Classification

Severity: SEV-2 (Major) - a critical business function (order processing) was completely unavailable.

Classified SEV-2 rather than SEV-1 because the failure was isolated to one service and did not take down the entire platform. Authentication, product browsing, and other features remained functional.

4. Timeline of Events

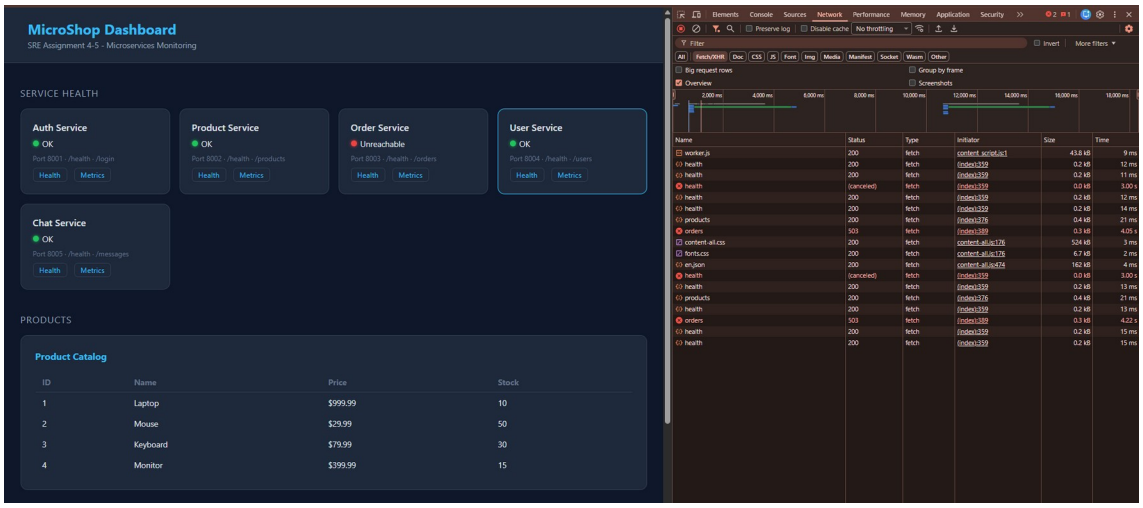
Time	Event
T+00:00	DB_HOST changed from 'postgres' to 'wrong-host' in docker-compose.yml
T+00:05	Order Service restarted: docker compose up -d --force-recreate order-service
T+00:10	Frontend dashboard detected Order Service as 'Unreachable'
T+00:15	/api/orders/orders returned HTTP 503 Service Unavailable
T+00:20	order_errors_total counter began incrementing in Prometheus
T+00:25	/health returned 503: 'DB connection failed: could not translate host name wrong-host'
T+00:30	docker logs confirmed repeated psycopg2 DNS resolution failures
T+01:00	Root cause confirmed: incorrect DB_HOST environment variable
T+01:05	DB_HOST corrected back to 'postgres' in docker-compose.yml
T+01:10	Order Service restarted with corrected configuration
T+01:15	/health returned HTTP 200: {status: ok, db: connected}
T+01:20	Frontend dashboard confirmed all services green - full recovery

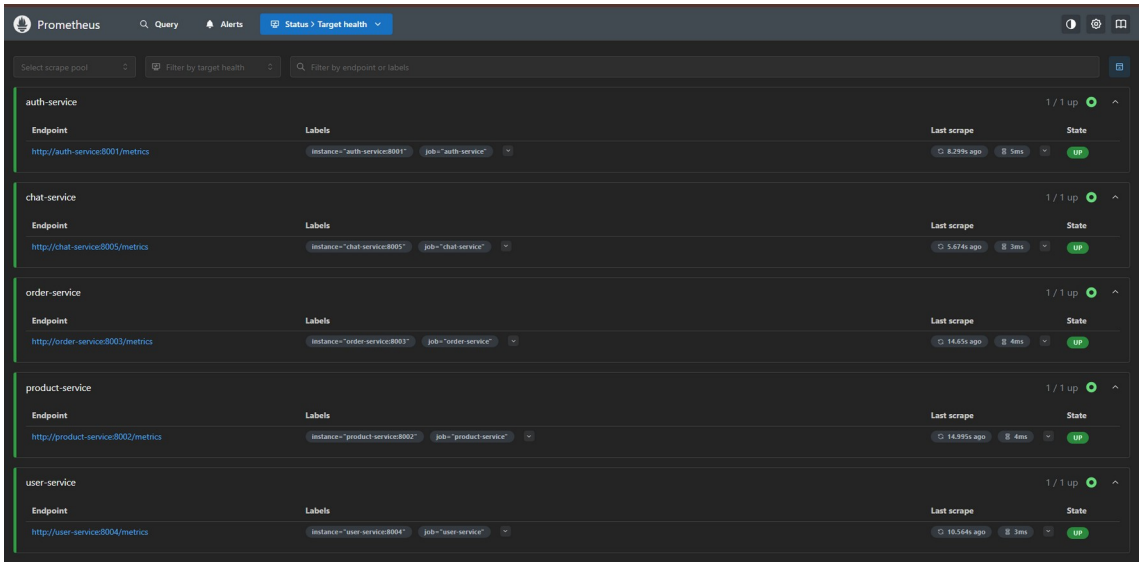
5. Root Cause Analysis

Root cause: misconfigured DB_HOST environment variable. Docker Compose services resolve each other by container name through Docker's internal DNS. The hostname 'postgres' resolves correctly on the microshop-net network because a container with that name exists. The value 'wrong-host' has no DNS record on the network, so every call to psycopg2.connect() immediately raised an OperationalError at the DNS resolution step - before even attempting a TCP connection to port 5432.

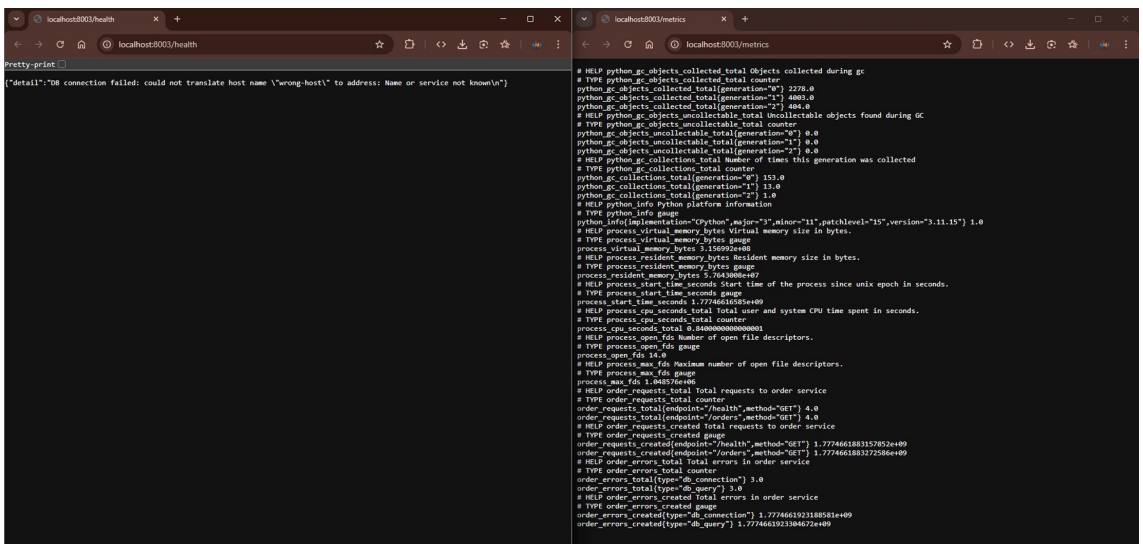
Key SRE observation: Prometheus showed order-service as up=1 throughout the entire incident. This is because Prometheus scrapes /metrics, which does not touch the database and always responds successfully. The service process was alive, but functionally broken. This demonstrates a critical distinction in SRE: process liveness does not equal service health. The /health endpoint (which calls psycopg2.connect() on every request) correctly returned 503 and provided the exact error message pointing to the bad hostname.

Incident Evidence





```
Nurzhan@Nurzhan MINGW64 ~/Favorites/Basa/Education/Programming/Introduction to SRE/assignment-4-5
$ docker logs order-service 2>&1 | tail -20
INFO:     172.19.0.3:52208 - "GET /orders HTTP/1.1" 503 Service Unavailable
INFO:     172.19.0.9:57344 - "GET /metrics HTTP/1.1" 200 OK
INFO:     172.19.0.3:52114 - "GET /orders HTTP/1.1" 503 Service Unavailable
INFO:     172.19.0.9:42486 - "GET /metrics HTTP/1.1" 200 OK
INFO:     172.19.0.3:43896 - "GET /orders HTTP/1.1" 503 Service Unavailable
INFO:     172.19.0.3:40350 - "GET /orders HTTP/1.1" 503 Service Unavailable
INFO:     172.19.0.9:34554 - "GET /metrics HTTP/1.1" 200 OK
INFO:     172.19.0.3:37070 - "GET /orders HTTP/1.1" 503 Service Unavailable
INFO:     172.19.0.9:42780 - "GET /metrics HTTP/1.1" 200 OK
INFO:     172.19.0.3:55036 - "GET /orders HTTP/1.1" 503 Service Unavailable
INFO:     172.19.0.3:51186 - "GET /orders HTTP/1.1" 503 Service Unavailable
INFO:     172.19.0.9:49238 - "GET /metrics HTTP/1.1" 200 OK
INFO:     172.19.0.3:54910 - "GET /orders HTTP/1.1" 503 Service Unavailable
INFO:     172.19.0.9:33888 - "GET /metrics HTTP/1.1" 200 OK
INFO:     172.19.0.3:58512 - "GET /orders HTTP/1.1" 503 Service Unavailable
INFO:     172.19.0.3:53038 - "GET /orders HTTP/1.1" 503 Service Unavailable
INFO:     172.19.0.9:34342 - "GET /metrics HTTP/1.1" 200 OK
INFO:     172.19.0.3:50634 - "GET /orders HTTP/1.1" 503 Service Unavailable
INFO:     172.19.0.9:55712 - "GET /metrics HTTP/1.1" 200 OK
INFO:     172.19.0.3:58176 - "GET /orders HTTP/1.1" 503 Service Unavailable

Nurzhan@Nurzhan MINGW64 ~/Favorites/Basa/Education/Programming/Introduction to SRE/assignment-4-5
$
```

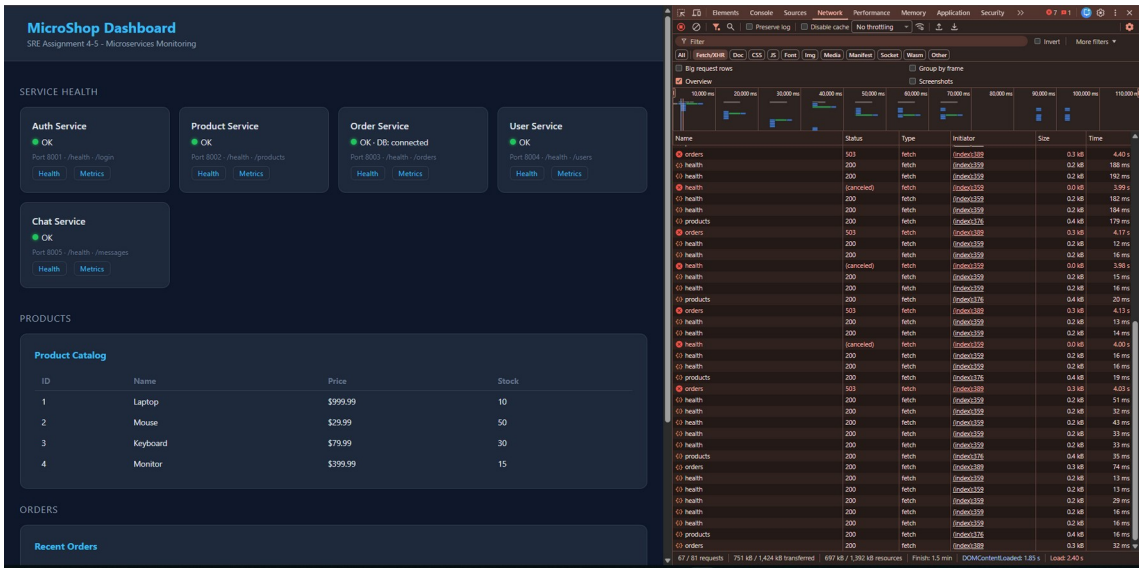


6. Mitigation Steps

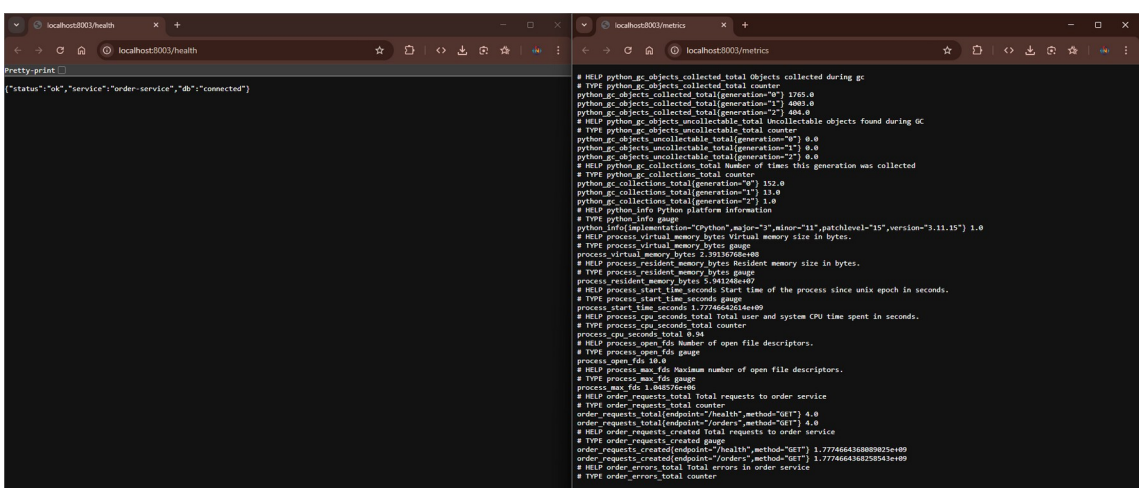
- Step 1 - Identified failing container: docker logs order-service showed repeated 503 on /orders
- Step 2 - Checked /health endpoint: returned exact error message with the bad hostname 'wrong-host'
- Step 3 - Inspected docker-compose.yml: found DB_HOST set to 'wrong-host' instead of 'postgres'
- Step 4 - Applied fix: changed DB_HOST back to 'postgres'
- Step 5 - Restarted only the affected container: docker compose up -d --force-recreate order-service
- Step 6 - Verified recovery: /health returned 200 with db: connected
- Step 7 - Confirmed via dashboard: all services showing green

7. Resolution Confirmation

Full recovery was achieved within approximately 70 seconds of applying the fix. The /health endpoint returned HTTP 200 with db: connected, the frontend dashboard showed all five services green, and the order_errors_total counter stopped incrementing in Prometheus. No data was lost and no other services required any intervention.




```
Nurzhan@Nurzhan MINGW64 ~/Favorites/Basa/Education/Programming/Introduction to SRE/assignment-4-5
$ docker logs order-service 2>&1 | tail -10
INFO:     172.19.0.3:41564 - "GET /orders HTTP/1.1" 200 OK
INFO:     172.19.0.9:46448 - "GET /metrics HTTP/1.1" 200 OK
INFO:     172.19.0.3:36050 - "GET /health HTTP/1.1" 200 OK
INFO:     172.19.0.3:36060 - "GET /orders HTTP/1.1" 200 OK
INFO:     172.19.0.9:56206 - "GET /metrics HTTP/1.1" 200 OK
INFO:     172.19.0.3:46088 - "GET /health HTTP/1.1" 200 OK
INFO:     172.19.0.3:46094 - "GET /orders HTTP/1.1" 200 OK
INFO:     172.19.0.3:54568 - "GET /health HTTP/1.1" 200 OK
INFO:     172.19.0.3:54578 - "GET /orders HTTP/1.1" 200 OK
INFO:     172.19.0.9:51814 - "GET /metrics HTTP/1.1" 200 OK

Nurzhan@Nurzhan MINGW64 ~/Favorites/Basa/Education/Programming/Introduction to SRE/assignment-4-5
```



Postmortem Analysis

1. Incident Overview

On 2026-04-29, the MicroShop Order Service became fully unavailable for approximately 70 seconds due to an incorrect database hostname in the service environment configuration. The incident was intentionally introduced as part of Assignment 4 to simulate a realistic production failure and practice the complete incident response lifecycle - detection, analysis, mitigation, and recovery.

2. Customer Impact

- 100% of order-related requests (view orders, create order) failed with HTTP 503
- The MicroShop frontend showed Order Service as 'Unreachable' for the duration of the incident
- No data loss occurred - PostgreSQL remained healthy and all existing orders were preserved
- Product browsing, authentication, user profiles, and chat were fully unaffected
- In a real production environment this would represent complete loss of revenue-generating functionality

3. Root Cause Analysis

The DB_HOST environment variable was manually changed to a non-existent hostname. In Docker Compose, services discover each other by container name via Docker's internal DNS. Setting DB_HOST='wrong-host' caused immediate DNS lookup failure on every database operation. The service had no connection retry logic - each incoming request attempted a fresh psycopg2.connect() with a 3-second timeout, failed instantly, incremented order_errors_total, and returned HTTP 503.

Contributing factor: Prometheus monitoring showed the service as UP throughout the incident because it only checks /metrics endpoint availability (process liveness), not actual DB connectivity (functional health). This created a misleading signal that could delay detection in a real environment without proper alerting on error rates.

4. Detection and Response Evaluation

Detection was fast - the frontend dashboard showed the failure within 10 seconds of the container restart. The /health endpoint was the critical diagnostic tool: it provided the exact error message including the bad hostname, making root cause identification immediate.

Gap identified: Prometheus showed up=1 during the incident. In production, relying solely on process liveness metrics would delay detection. The OrderServiceErrors alert rule added to alert_rules.yml addresses this - it fires when order_errors_total increases, catching functional failures that process liveness cannot detect.

5. Resolution Summary

A single configuration change (correcting DB_HOST to 'postgres') followed by a container restart fully restored the service in under 2 minutes. No code changes, no data recovery, and no other service required any intervention. The microservices fault isolation architecture worked exactly as designed.

6. Lessons Learned

- Environment variables are a frequent source of misconfiguration - they should be validated at service startup
- Process liveness (up=1 in Prometheus) does not mean a service is functionally healthy
- Dependency health checks in /health endpoints are essential - they reveal failures that Prometheus scraping misses
- Microservices fault isolation worked correctly - a broken Order Service did not cascade to other services
- The /health endpoint design (checking actual DB connectivity on every call) was the correct approach and proved essential
- Manual changes to running configurations should always be validated before applying in production

7. Action Items

- DONE - Added Prometheus alert rules: OrderServiceErrors and ServiceDown in monitoring/alert_rules.yml
- DONE - Grafana dashboard auto-provisioned with error rate panels visible during incident
- DONE - Resource limits and Docker healthchecks added to all containers
- TODO - Add DB_HOST validation at Order Service startup: attempt connection and fail fast with clear error
- TODO - Implement connection retry with exponential backoff in psycopg2 calls
- TODO - Configure Grafana alert notification channel (email/Slack) connected to Prometheus alerts
- TODO - Replace 0.0.0.0/0 in Terraform security group with specific IP ranges in production

ASSIGNMENT 5 - Infrastructure as Code (Terraform + AWS)

Overview

The cloud infrastructure for MicroShop was provisioned using Terraform with the AWS provider. Terraform is a declarative IaC tool - the entire infrastructure is described in .tf files and created, updated, or destroyed with single commands. This makes infrastructure reproducible, version-controlled, and auditable - unlike manual AWS Console operations.

Implementation: Two Phases

During development, Terraform was first configured with a local Docker provider (kreuzwerker/docker) to validate the configuration structure locally. This phase is visible in screenshots 13_terraform_apply_*. After the professor confirmed that cloud deployment was required, the configuration was migrated to the AWS provider. The final AWS deployment is shown in screenshots 14_terraform_apply_after_AWS_implementation_*. The code in the repository reflects the final AWS configuration only.

> Note: The AWS apply created: EC2 instance at 34.238.137.187 (us-east-1), security group sg-0091acf407323802f. The Docker provider files were removed from state before the AWS migration.

Terraform Files

main.tf

Defines two AWS resources. aws_security_group creates a firewall with inbound rules for SSH (22), HTTP (80), Grafana (3000), and Prometheus (9090). SSH and monitoring ports use the allowed_ssh_cidr variable - set it to your IP in production to restrict access. aws_instance provisions a t3.micro EC2 VM with Amazon Linux 2. The user_data script runs on first boot and installs Docker, git, and docker-compose so the application can be deployed immediately after the instance starts. The key_name field accepts an AWS key pair name via the ssh_key_name variable - left empty for this demo but required for SSH access in a real deployment.

variables.tf

Declares five input variables: aws_region, instance_type, ami_id, allowed_ssh_cidr, and ssh_key_name. Using variables instead of hardcoded values makes the configuration reusable - changing the region, instance size, or IP restrictions requires editing only terraform.tfvars, not the core infrastructure code.

terraform.tfvars

Provides actual values: us-east-1, t3.micro, Amazon Linux 2 AMI. t3.micro was chosen because t2.micro was rejected by AWS as not free-tier eligible in the selected region. Includes comments explaining how to restrict access in production.

outputs.tf

Defines what Terraform prints after apply: instance_public_ip, instance_public_dns, grafana_url, prometheus_url, and security_group_id. These make it easy to access deployed services without looking up the IP manually in the AWS Console.

Security Considerations

- SSH and monitoring ports use allowed_ssh_cidr variable - defaults to 0.0.0.0/0 for demo, restrict to specific IP in production
- No SSH key pair configured by default - set ssh_key_name variable to an existing AWS key pair to enable SSH access
- Grafana and Prometheus have no built-in auth when exposed publicly - in production place behind a VPN or add authentication
- terraform.tfstate is excluded from git via .gitignore - it contains AWS resource IDs and should never be committed

Deployment Commands

- terraform init - downloads hashicorp/aws provider plugin, initializes working directory
- terraform plan - previews all changes; shows exact resources to be created before any AWS calls
- terraform apply - creates resources on AWS (completed in ~30 seconds)
- terraform destroy - removes all resources cleanly; always run after assignment to avoid AWS charges

Terraform Screenshots

```
Nurzhan@Nurzhan MINGW64 ~/Favorites/Basa/Education/Programming/Introduction to SRE/assignment-4-5/terraform
$ cd ~/Favorites/Basa/Education/Programming/Introduction\ to\ SRE/assignment-4-5/terraform
terraform init
Initializing the backend...
Initializing provider plugins...
- Reusing previous version of kreuzwerker/docker from the dependency lock file
- Finding hashicorp/aws versions matching "~> 5.0"...
- Using previously-installed kreuzwerker/docker v3.9.0
- Installing hashicorp/aws v5.100.0...
- Installed hashicorp/aws v5.100.0 (signed by HashiCorp)
Terraform has made some changes to the provider dependency selections recorded
in the .terraform.lock.hcl file. Review those changes and commit them to your
version control system if they represent changes you intended to make.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.

Nurzhan@Nurzhan MINGW64 ~/Favorites/Basa/Education/Programming/Introduction to SRE/assignment-4-5/terraform
$ []
```

```
Nurzhan@Nurzhan MINGW64 ~/Favorites/Basa/Education/Programming/Introduction to SRE/assignment-4-5/terraform
$ mkdir -p ~/.aws
cat > ~/.aws/credentials << 'EOF'
[default]
aws_access_key_id = AKIAUCXIRKVEJNVBJZ4V
aws_secret_access_key = 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
EOF

cat > ~/.aws/config << 'EOF'
[default]
region = us-east-1
EOF
```

PROBLEMS DEBUG CONSOLE OUTPUT TERMINAL PORTS

```
Nurzhan@Nurzhan MINGW64 ~/Favorites/Basa/Education/Programming/Introduction to SRE/assignment-4-5/terraform
$ terraform plan
docker_image.prometheus: Refreshing state... [id=sha256:dda13e28bf95a5e5ca5b8ed56852006094c1c8e8871d9c9dbeed30aa6e55271fprom/prometheus:latest]
docker_image.grafana: Refreshing state... [id=sha256:83749231c3835e390a3144e5e940203e42b9589761f20ef3169c716e734ad505grafana/grafana:latest]
docker_image.postgres: Refreshing state... [id=sha256:09e4f20b14ddb3dfe3a0c825b206032aaf8f28300ba2070c0b60fc1c10c6abc7postgres:15-alpine]
docker_network.microshop_net: Refreshing state... [id=cc7a9373c9a0f0077fb33e635d671dbaa0a62aaf7624daf7fb7f2a194fed1760]
docker_container.prometheus: Refreshing state... [id=c5a4126c07e2ca3093263116a604aedd117240c79150e437db2cafb5d97361ca]
docker_container.postgres: Refreshing state... [id=77cb9333a38a11f265369538b5ef766f7f266f0fb26f5e6ab8ebf8b32f5e6507]
docker_container.grafana: Refreshing state... [id=39ee0065e388d3e4adb07ca613de55db7a85f5825f09617d1c7dbe531179ca67]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform planned the following actions, but then encountered a problem:

  # aws_instance.microshop will be created
  + resource "aws_instance" "microshop" {
      + ami                                  = "ami-0c02fb55956c7d316"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + cpu_core_count                       = (known after apply)
      + cpu_threads_per_core                 = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t2.micro"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = (known after apply)
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
      + password_data                        = (known after apply)
      + placement_group                      = (known after apply)
      + placement_partition_number           = (known after apply)
      + primary_network_interface_id         = (known after apply)
      + private_dns                          = (known after apply)
      + private_ip                           = (known after apply)
      + public_dns                           = (known after apply)
      + public_ip                            = (known after apply)
      + secondary_private_ips                = (known after apply)
      + security_groups                      = (known after apply)
```

PROBLEMS DEBUG CONSOLE OUTPUT TERMINAL PORTS

```
      + security_groups                      = (known after apply)
      + source_dest_check                    = true
      + spot_instance_request_id             = (known after apply)
      + subnet_id                            = (known after apply)
      + tags                                 = {
          + "Name" = "microshop-sre-instance"
        }
      + tags_all                             = {
          + "Name" = "microshop-sre-instance"
        }
      + tenancy                              = (known after apply)
      + user_data                            = "2faa7d128b9ce78d681934d1702d56c609b74c55"
      + user_data_base64                     = (known after apply)
      + user_data_replace_on_change          = false
      + vpc_security_group_ids               = (known after apply)

      + capacity_reservation_specification (known after apply)

      + cpu_options (known after apply)

      + ebs_block_device (known after apply)

      + enclave_options (known after apply)

      + ephemeral_block_device (known after apply)

      + instance_market_options (known after apply)

      + maintenance_options (known after apply)

      + metadata_options (known after apply)

      + network_interface (known after apply)

      + private_dns_name_options (known after apply)

      + root_block_device (known after apply)
    }

  # aws_security_group.microshop_sg will be created
  + resource "aws_security_group" "microshop_sg" {
      + arn                    = (known after apply)
      + description            = "Security group for MicroShop SRE assignment"
      + egress                 = [
          + {
              + cidr_blocks      = [
                  + "0.0.0.0/0",
                ]
              + from_port        = 0
              + ipv6_cidr_blocks = []
              + prefix_list_ids  = []
```

```
Nurzhan@Nurzhan MINGW64 ~/Favorites/Basa/Education/Programming/Introduction to SRE/assignment-4-5/terraform
$ terraform apply -auto-approve
aws_security_group.microshop_sg: Refreshing state... [id=sg-0091acf407323802f]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_instance.microshop will be created
  + resource "aws_instance" "microshop" {
      + ami                                  = "ami-0c02fb55956c7d316"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + cpu_core_count                       = (known after apply)
      + cpu_threads_per_core                 = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t3.micro"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = (known after apply)
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
      + password_data                        = (known after apply)
      + placement_group                      = (known after apply)
      + placement_partition_number           = (known after apply)
      + primary_network_interface_id         = (known after apply)
      + private_dns                          = (known after apply)
      + private_ip                           = (known after apply)
      + public_dns                           = (known after apply)
      + public_ip                            = (known after apply)
      + secondary_private_ips                = (known after apply)
      + security_groups                      = (known after apply)
      + source_dest_check                    = true
      + spot_instance_request_id             = (known after apply)
      + subnet_id                            = (known after apply)
      + tags                                 = {
          + "Name" = "microshop-sre-instance"
```

```
Changes to Outputs:
  ~ grafana_url         = "http://localhost:3001" -> (known after apply)
  + instance_public_dns = (known after apply)
  + instance_public_ip  = (known after apply)
  ~ prometheus_url      = "http://localhost:9091" -> (known after apply)
aws_instance.microshop: Creating...
aws_instance.microshop: Still creating... [00m10s elapsed]
aws_instance.microshop: Creation complete after 15s [id=i-0123ea19e003650eb]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

grafana_url = "http://34.238.137.187:3000"
instance_public_dns = "ec2-34-238-137-187.compute-1.amazonaws.com"
instance_public_ip = "34.238.137.187"
prometheus_url = "http://34.238.137.187:9090"
security_group_id = "sg-0091acf407323802f"

Nurzhan@Nurzhan MINGW64 ~/Favorites/Basa/Education/Programming/Introduction to SRE/assignment-4-5/terraform
$ 
```

PROBLEMS DEBUG CONSOLE OUTPUT TERMINAL PORTS

```
      + spot_instance_request_id             = (known after apply)
      + subnet_id                            = (known after apply)
      + tags                                 = {
          + "Name" = "microshop-sre-instance"
        }
      + tags_all                             = {
          + "Name" = "microshop-sre-instance"
        }
      + tenancy                              = (known after apply)
      + user_data                            = "2faa7d128b9ce78d681934d1702d56c609b74c55"
      + user_data_base64                     = (known after apply)
      + user_data_replace_on_change          = false
      + vpc_security_group_ids               = [
          + "sg-0091acf407323802f",
        ]

      + capacity_reservation_specification (known after apply)

      + cpu_options (known after apply)

      + ebs_block_device (known after apply)

      + enclave_options (known after apply)

      + ephemeral_block_device (known after apply)

      + instance_market_options (known after apply)

      + maintenance_options (known after apply)

      + metadata_options (known after apply)

      + network_interface (known after apply)

      + private_dns_name_options (known after apply)

      + root_block_device (known after apply)
    }

Plan: 1 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  ~ grafana_url         = "http://localhost:3001" -> (known after apply)
  + instance_public_dns = (known after apply)
  + instance_public_ip  = (known after apply)
  ~ prometheus_url      = "http://localhost:9091" -> (known after apply)
aws_instance.microshop: Creating...
aws_instance.microshop: Still creating... [00m10s elapsed]
aws_instance.microshop: Creation complete after 15s [id=i-0123ea19e003650eb]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
```

Why Terraform over Manual AWS Console

- Reproducible - running terraform apply again produces identical infrastructure every time
- Version controlled - infrastructure changes tracked in git alongside application code
- Self-documenting - .tf files describe exactly what exists and why
- Automated - no manual clicking, no missed steps, no human error
- Destroyable - terraform destroy cleanly removes everything, preventing surprise AWS charges

Deployment Guide

Prerequisites

- Docker Desktop installed and running
- Terraform >= 1.0 installed
- AWS account with IAM user (AdministratorAccess policy, access key + secret key)
- Git

1. Clone the Repository

```
git clone https://github.com/Nurlybek-Nurzhan/Assignment_4-5_SRE.git
cd Assignment_4-5_SRE
```

2. Configure Environment

```
cp .env.example .env
```

Edit .env and set your preferred passwords and JWT secret.

3. Run Locally with Docker Compose

```
docker compose up --build
```

Dashboard: http://localhost:80 | Prometheus: http://localhost:9090 | Grafana: http://localhost:3000

Login with credentials from .env (default: nurzhan / password123).

4. Provision AWS Infrastructure

```
cd terraform
aws configure   # enter IAM access key, secret key, region: us-east-1
terraform init && terraform plan && terraform apply
```

SSH into the instance, clone the repo, and run docker compose up.

5. Simulate the Incident

In docker-compose.yml change DB_HOST: postgres → DB_HOST: wrong-host

docker compose up -d --force-recreate order-service

Observe Order Service become Unreachable. Check /health for the error message. Watch order_errors_total in Prometheus.

6. Recover from the Incident

Restore DB_HOST: postgres in docker-compose.yml

docker compose up -d --force-recreate order-service

All services return to healthy state within ~10 seconds.

7. Clean Up AWS Resources

cd terraform && terraform destroy

Always run this after the assignment to avoid ongoing AWS charges.